Memory aware SCADE

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Objectives

Drive memory use in a purely functional (data-flow) language

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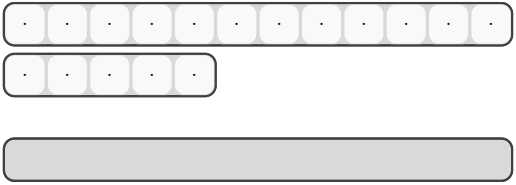

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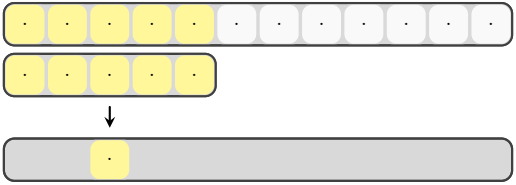

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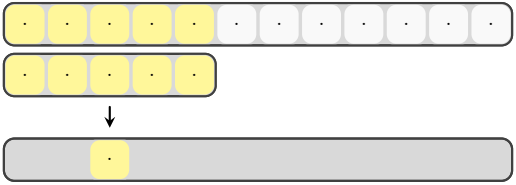

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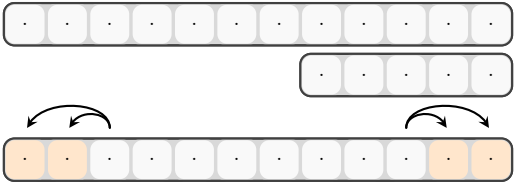

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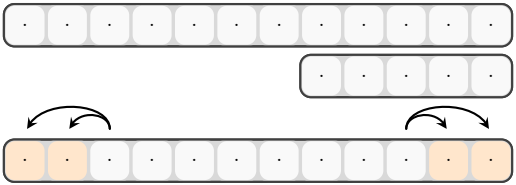

[1] J-L Gaudiot et al. "The Sisal model of functional programming and its implementation". In: *Proceedings of IEEE International Symposium on Parallel Algorithms Architecture Synthesis*. IEEE. 1997, pp. 112–123.

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Calling imperative code: safely encapsulate argument modification

... while preserving

- the data-flow style (fully parallel)
- referential transparency

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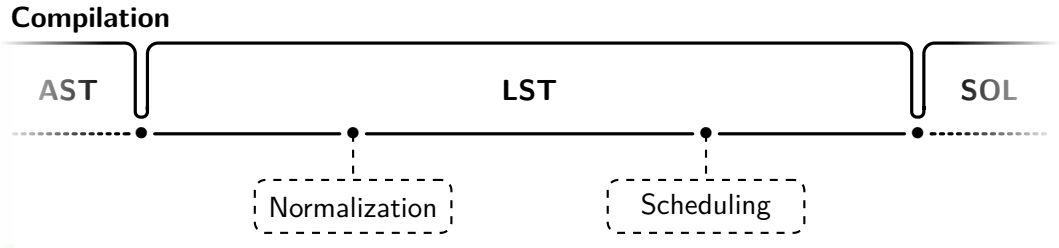
Compiler middle-end



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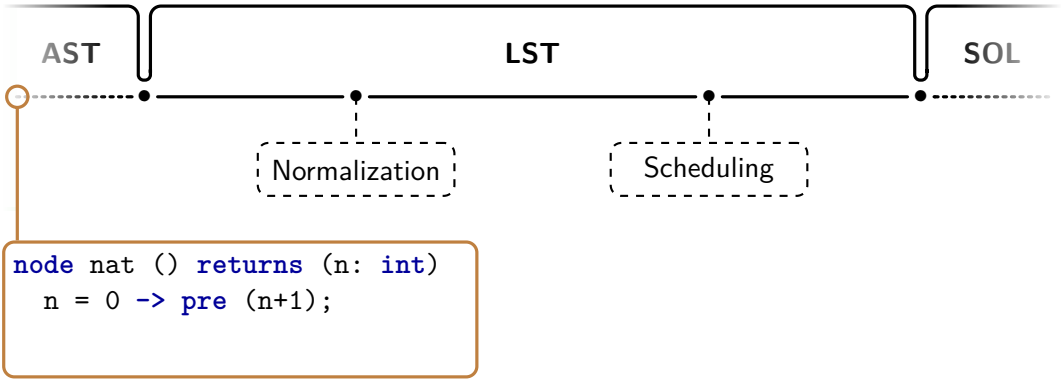

- Referential transparency
- Data-flow representation

⇒ **Implicit copies**

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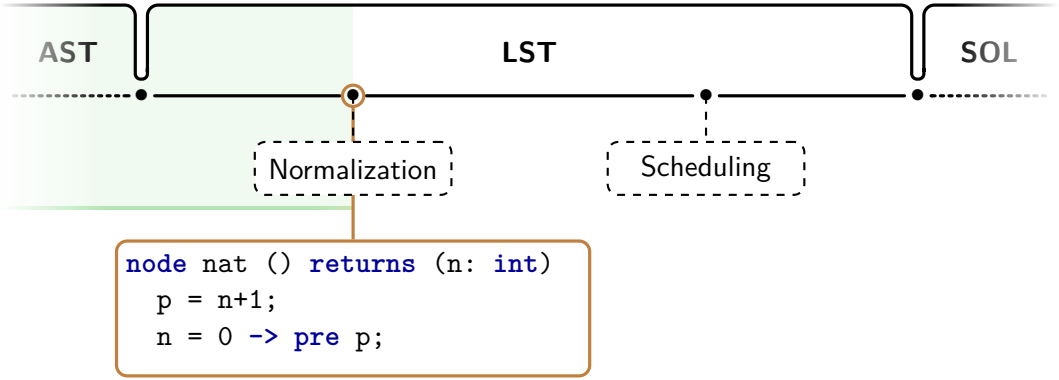

- ~~Referential transparency~~
- Data-flow representation

⇒ **Identified memory (state)**

Compiler middle-end

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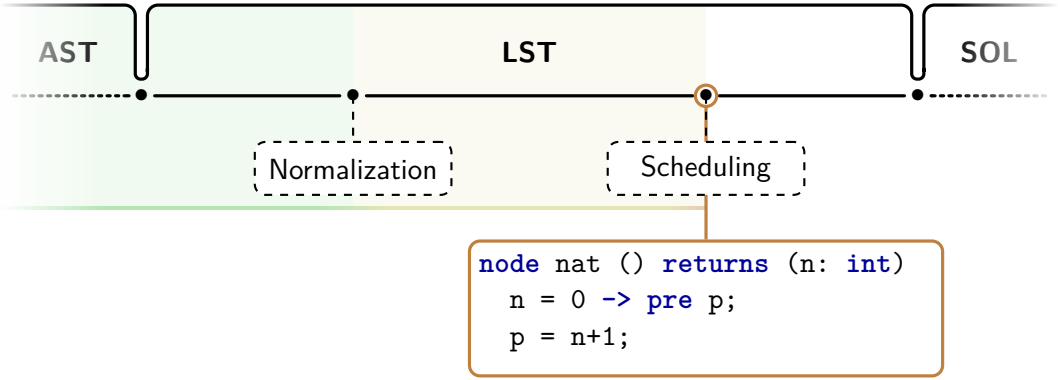

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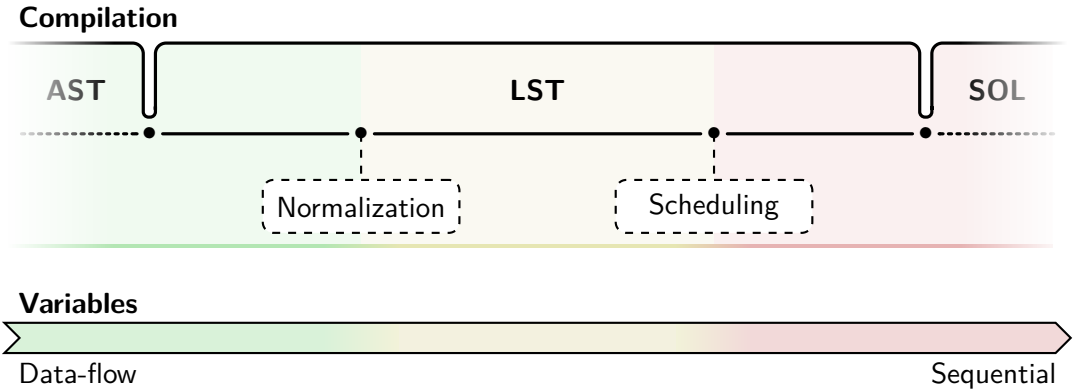

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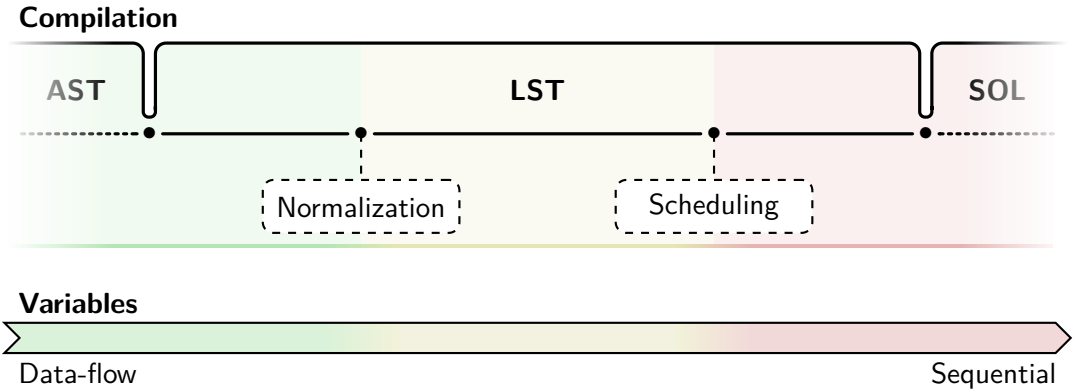
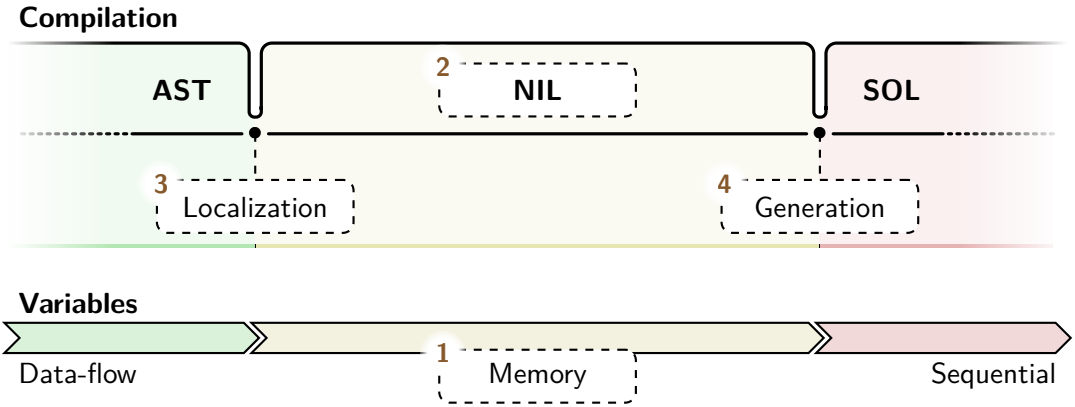


Agenda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AST
2
NIL
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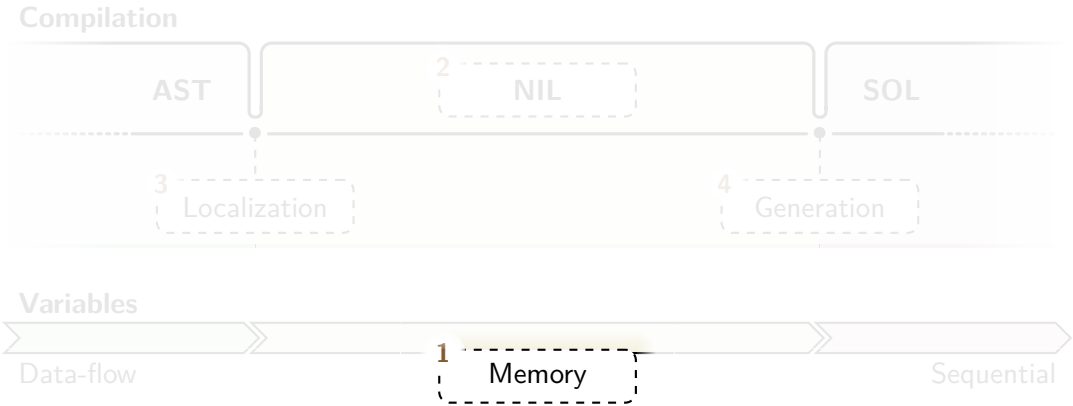
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What's in memory ?

Structured data

τ	$::=$		***Types***
	$\mid$	$\alpha,\ \beta,\ \gamma$	variable
	$\mid$	$\texttt{int} \mid \texttt{bool}$	base types
	$\mid$	$\tau * \ldots * \tau$	tuple
	$\mid$	$[n]\tau$	array

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Memory location

$$\varepsilon.f$$

Typed memory variable

Projection function

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Intuition: Viewing (part of) structured data differently by mapping accesses

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Array transposition:

$$[n]\,[p]\,\tau \Longleftarrow \Big[\texttt{[j<}n\texttt{]}\ \texttt{[i<}p\texttt{]} \leftarrow \texttt{[i<}p\texttt{]}\ \texttt{[j<}n\texttt{]}\Big] \Longrightarrow [p]\,[n]\,\tau$$

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Array splitting:

$$[n+p]\,\tau \Longrightarrow \begin{bmatrix} \texttt{[i+0<}n+p\texttt{]} & \leftarrow & \texttt{.1}/2\ \texttt{[i<}n\texttt{]} \\ \texttt{[i+n<}n+p\texttt{]} & \leftarrow & \texttt{.2}/2\ \texttt{[i<}p\texttt{]} \end{bmatrix} \Longrightarrow [n]\,\tau * [p]\,\tau$$

Operations on projection functions

Properties

- Injective $\Longrightarrow$ ✓ Editing in place

slices

$$[n+p-1]\,\alpha = \left[\texttt{[i+j<}n+p-1\texttt{]} \;\leftarrow\; \texttt{[i<}n\texttt{]}\;\texttt{[j<}p\texttt{]}\right] \Longrightarrow [n]\,[p]\,\alpha$$

Operations on projection functions

Properties

- Injective $\Longrightarrow$ ✓ Editing in place
- Single piece $\Longrightarrow$ ✓ Passing by pointer

---

map snd

$$[n](\alpha * \beta) = \left[\texttt{[i<}n\texttt{]}\ \texttt{.2}/2 \leftarrow \texttt{[i<}n\texttt{]}\right] \Longrightarrow [n]\beta$$

Operations on projection functions

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- Inclusion
- Disjunction

$\Longrightarrow$ Dependencies: $\cap\, p_i \setminus \cup q_i$

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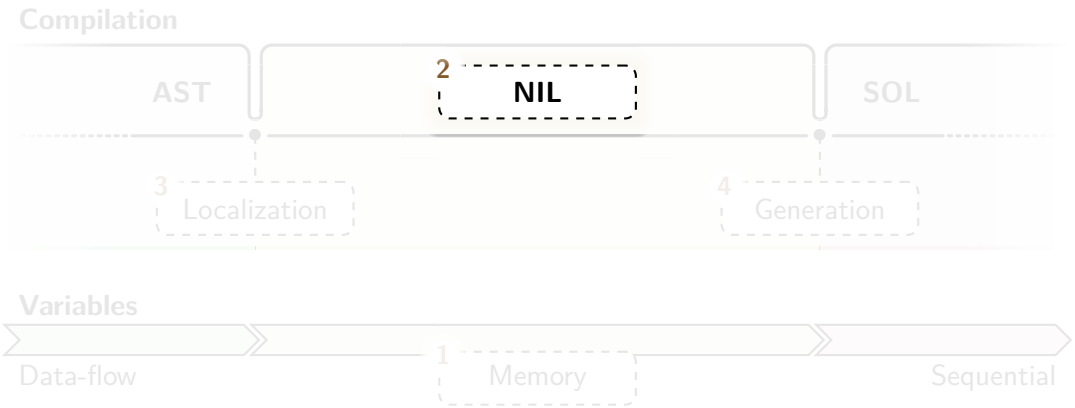
Transformations

- Composition
- Normal form (simplification)

$\Longrightarrow$ $\mathtt{tr}_{[n][p]\alpha} \circ \mathtt{tr}_{[p][n]\alpha} = I_{[n][p]\alpha}$

Agenda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No-copy Intermediate Language

No-copy Intermediate Language

e	$::=$		***Expressions***
	$\mid$	x	variable
	$\mid$	n	constant
	$\mid$	$e + e \mid \ldots$	operators
	$\mid$	$x\ (e, \ldots, e)$	instanciation
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Static verification

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$\Gamma \vdash e : \tau$

Memory location

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- No principal memory locations
- Semantics is independent of memory locations
- Locations filters out the set valid schedules

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Interfaces $\forall \varepsilon_1 : [n + p]\tau,\ \varepsilon_2 : [p]\tau,\ (\varepsilon_1.[],\ \varepsilon_2.[]) \to (\varepsilon_1.[]) \mid \varepsilon_1.[]$

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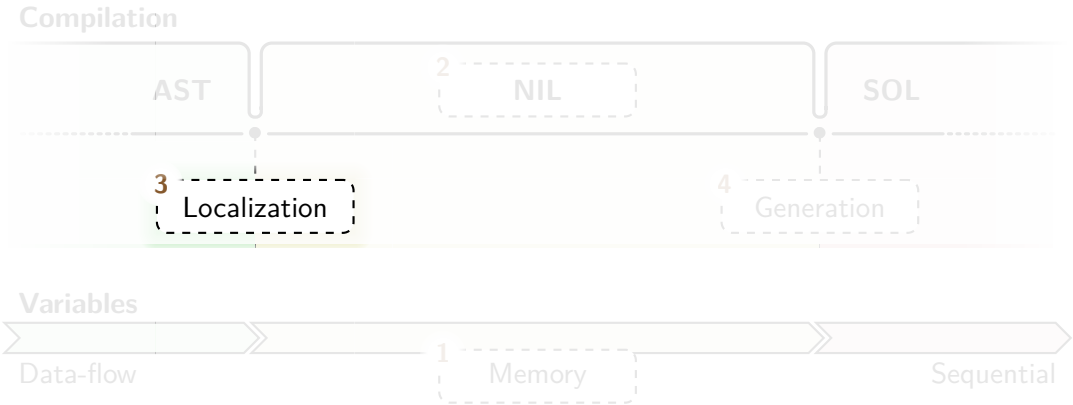
- No alias for modified locations (Scheduling)

Agenda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Location inference

Projection function unification: very under-constrained problem

- Restricted to simple unification (not inferring projection functions)
- Arrays and tuples build *single-piece* data

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Errors

- ✗ Inconsistent locations
- ✗ Overlapping copies
- ✗ Undeclared sharing in interface (recursion)
- ✗ Non single-piece arguments (instantiation)

Dependencies analysis & scheduling

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let swap (p) = (q at p)
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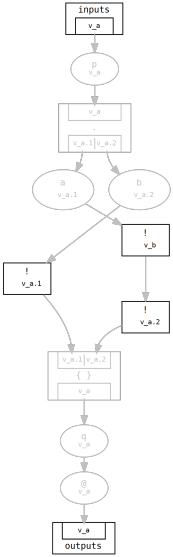
   1.1 Data-flow construction



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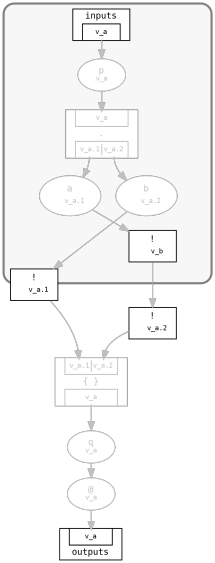

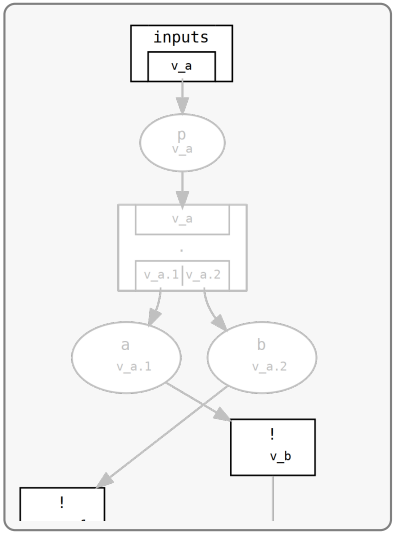

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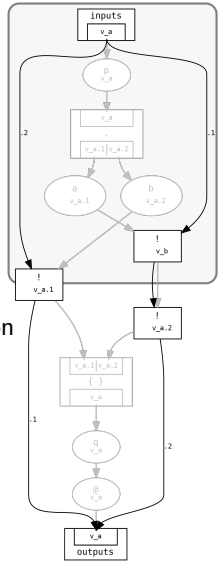

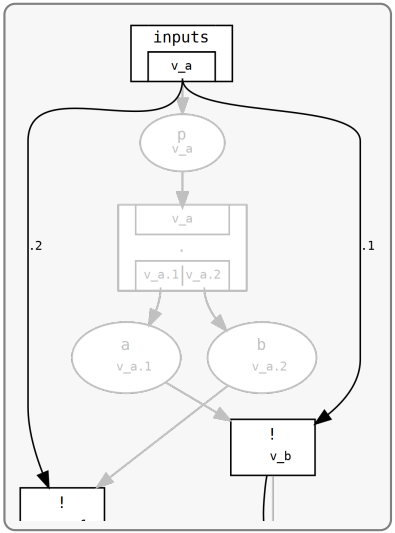

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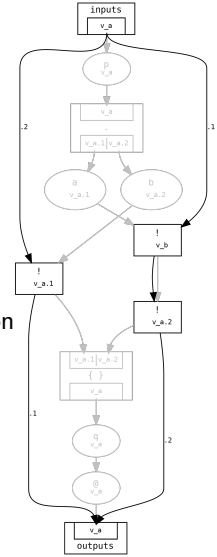

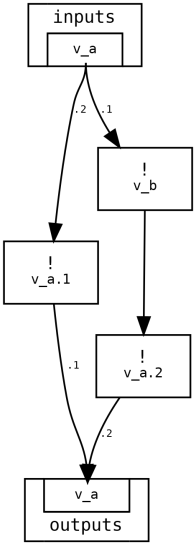

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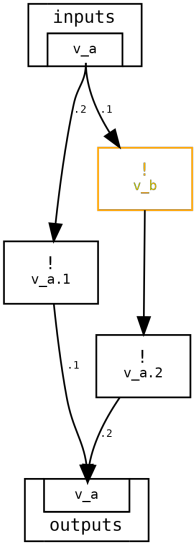

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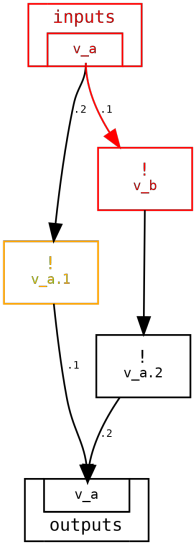

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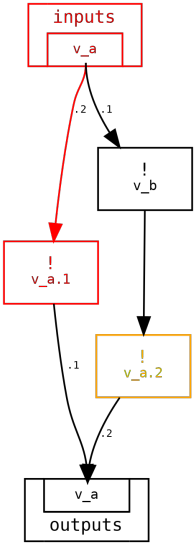

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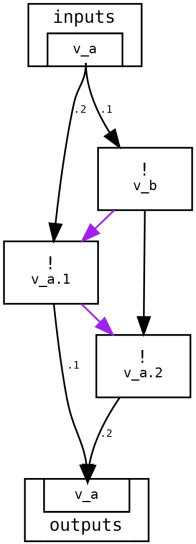
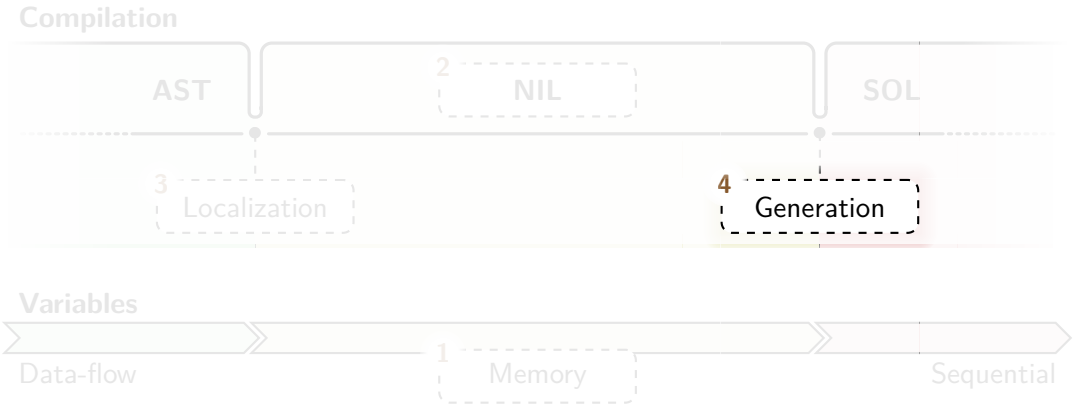


Agenda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Data representation

- Raw data for memory variables (no indirections)
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Copy implementation: concretizing projection functions

- Simple assignment for scalar values
- `memcpy` for single-piece locations
- Multiple statements (loops) for other cases.

The `step` function

Calling conventions:

- Optional state (pointer)
- Pointers on memory variables by default
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Demo

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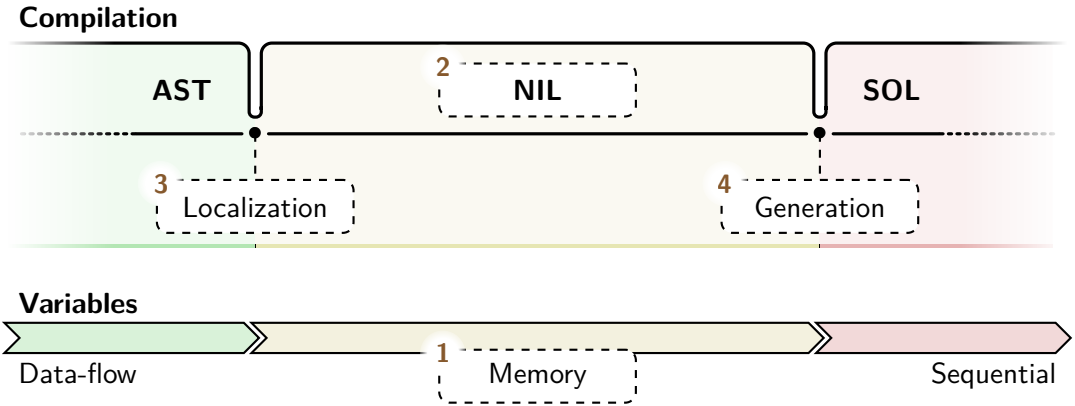
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What about SCADE?

Language: a conservative extension

- Implicit copies: interface, data-construction
- Copy-less assignments and projections (`x = y.1;`)
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- Array combinators for index functions (concatenation, transposition, ...)
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Extended causality check: semi-linearity[2]

- Location may be modified at most once
- Simpler & more precise diagnostics
- Sufficient condition for scheduling

[2] Léonard Gérard et al. "A Modular Memory Optimization for Synchronous Data-Flow Languages. Application to Arrays in a Lustre Compiler". In: *Languages, Compilers and Tools for Embedded Systems (LCTES'12)*. Best paper award. ACM. Beijing, 2012. URL: `lctes12.pdf`.

Take away

A fully parallel imperative language

- Data-flow representation in a pure SSA form

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Thank you!

Will linear types change the world?[3]

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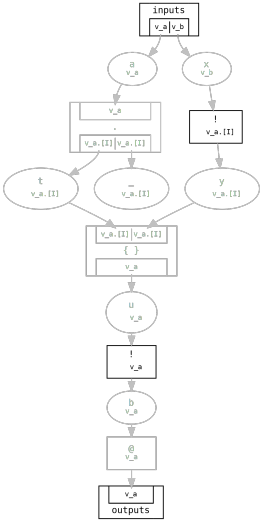

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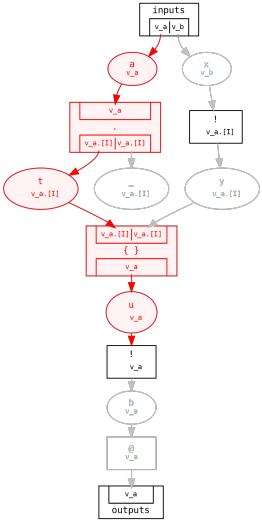

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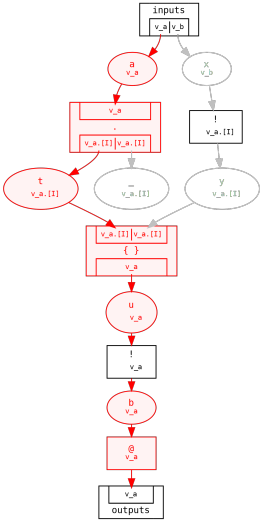

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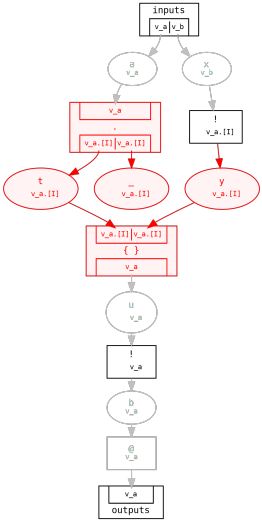

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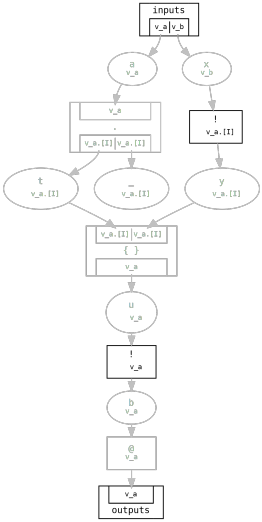

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